

Biases and ethics

- ✧ Genomic and pathway databases ingest the scientific literature
- ✧ The relative volume of literature in various fields is driven by funding
 - ✧ Databases are biased to cancer, heart disease and diabetes
 - ✧ Good luck if you have a rarer disease to treat



Biases and ethics

- ✦ Historically, minorities had lower chances of getting loans, mortgages, etc.
- ✦ This is reflected in the historical data
- ✦ This data is used to train ML models
- ✦ Now the historical racial bias is codified, since the models will make predictions based on patterns it saw in the training data
- ✦ Similar issue with clinical trials
 - ✦ Most studies were done on middle aged white men
 - ✦ Do these data apply to women, children, other ethnicities?
- ✦ Huge issue recognized in data ethics

